Classifying Probability Distributions via Statistical Fingerprinting: Training on Quantum Circuits, Testing on Real-World Data

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***Abstract*—We present a framework for classifying probability distributions using classical statistical features, motivated by the observation that many real-world processes generate observable outcomes whose distributional structure encodes meaningful information about the underlying generative process. Quantum circuit families—Instantaneous Quantum Polynomial (IQP), Clifford, and Clifford+T circuits—serve as canonical, reproducible sources of distinct distribution classes, providing labeled training data with known statistical structure. A suite of distribution statistics with polynomial shot requirements (Hamming-weight histogram, single-qubit marginals, parity bias, and nearest-neighbor connected ZZ correlations) is extracted from simulated circuit samples and used to train four classical machine-learning classifiers. We conduct a systematic scaling study over 4–20 qubits using a shot budget that scales quadratically with qubit count ($s = 16n^2$), yielding $N_c = 1000$ circuits per family. We show that two statistics commonly used for distribution characterization—Shannon entropy and collision probability—require exponential or sub-exponential shot budgets to estimate accurately and are therefore excluded in favor of features whose estimation cost scales polynomially with system size. A cross-domain experiment then demonstrates that classifiers trained exclusively on circuit-sampled data successfully classify aggregated real-world observations drawn from the same underlying distribution classes, with linear classifiers generalizing reliably while tree-based methods do not. These results establish a general pipeline: identify a real-world quantity as belonging to a known distribution class, generate labeled training data from that class using quantum circuits or classical simulation, and classify new observations without requiring labeled real-world examples.**
[TODO: Summarize key results: best accuracy, scaling trend.]

***Index Terms*—probability distribution classification, quantum circuit families, IQP circuits, Clifford circuits, statistical fingerprinting, quantum machine learning, near-term quantum computing**

I. Introduction

A recurring challenge across scientific and engineering domains is the classification of observations according to the underlying process that generated them. When the generative process is complex or partially unknown, direct model-based classification is infeasible. A distribution-fingerprinting approach offers an alternative: extract robust statistical features from observed samples, and classify by comparing those features to a known library of distribution signatures.

The practical utility of this approach depends on two capabilities. First, one needs a reliable method to characterize distributions with limited samples—that is, to extract features whose estimation cost scales favourably with problem size. Second, one needs labeled training data representative of the target distribution classes, which may be expensive or impossible to obtain directly from the real-world system of interest.

Quantum circuit families provide a principled solution to the second challenge. Different families—IQP, Clifford, and Clifford+T circuits—generate measurement-outcome distributions with structurally distinct statistical signatures. Crucially, these distributions can be generated in large quantities via classical simulation, providing abundant labeled training data for any classification pipeline. If a real-world process can be identified as generating data that resembles one of these distribution classes, classifiers trained on circuit data can be directly applied to real-world observations without requiring labeled field data.

In this work we ask: *can classical statistical features derived from quantum circuit measurement outcomes reliably classify distribution families, and do classifiers trained on circuit data transfer to aggregated real-world observations drawn from the same distribution classes?* A positive answer establishes the general utility of the circuit-as-distribution-generator paradigm and provides a concrete cross-domain validation methodology.

A key design decision concerns which statistical features to extract. Not all features are equally tractable: Shannon entropy and collision probability require exponentially or sub-exponentially many shots to estimate accurately as system size grows, which would demand exponentially growing measurement resources. We therefore restrict our feature set to statistics whose estimation cost scales polynomially with n, and adopt a shot budget that grows quadratically with qubit count to maintain consistent statistical quality across the full 4–20 qubit range.

Contributions:

- A framework that positions quantum circuit families as canonical, reproducible sources of labeled training data for distribution classification, enabling cross-domain generalization to real-world observations.
- A reproducible data-generation pipeline for three circuit families (IQP, Clifford, Clifford+T) spanning 4–20 qubits with a principled quadratic shot scaling law.
- An analysis of the shot requirements of each candidate feature, motivating a feature set restricted to statistics with polynomial estimation cost.
- A feature-extraction suite of four distributional statistics—Hamming-weight histogram, single-qubit marginals, parity bias, and connected ZZ correlations—that are agnostic to system size and flatten into a fixed-length vector of size $3n + 1$.
- A comparative evaluation of four classical classifiers (Logistic Regression, Decision Tree, Gradient Boosting, Extra Trees) across qubit counts.
- A cross-domain experiment demonstrating that Logistic Regression generalizes from circuit training data to aggregated real-world observations, revealing that the linear structure of the feature space is preserved under aggregation.

II. Background

A. Circuit Families as Distribution Classes

The central premise of our framework is that quantum circuit families define distinct, identifiable classes of probability distributions over binary strings. We consider three families that span a range of computational complexity and distributional structure.

a) Clifford circuits: Clifford circuits are generated by the gate set $\{H, S, \mathrm{CNOT}\}$ and are efficiently simulable classically via the stabilizer formalism [**?**]. Their output distributions are not uniformly random; they possess structured correlations that reflect the underlying stabilizer group.

b) Clifford+T circuits: Adding the $T = \mathrm{diag}(1, e^{i\pi/4})$ gate to the Clifford set yields a universal gate set [**?**]. The T gate breaks the Clifford structure and, with sufficient T-count, moves the circuit outside the efficiently simulable regime, altering the distributional signature in measurable ways.

c) IQP circuits: Instantaneous Quantum Polynomial (IQP) circuits apply only diagonal gates in the computational basis, sandwiched by Hadamard layers [**?**]. Classical hardness of sampling from IQP circuits is supported by complexity-theoretic arguments [**?**], and their parity structure produces a characteristic distributional fingerprint that is qualitatively distinct from the Clifford and Clifford+T families.

B. Classical Shadows and Distribution Statistics

[TODO: Brief review of classical shadows [?] and how our approach relates to (but differs from) full shadow tomography.]

C. Distribution Classification

[TODO: Brief review of relevant prior work on distribution testing, hypothesis testing between quantum distributions, and any classical analogs (e.g., signal classification by distributional features). Position our contribution relative to these.]

III. Methods

A. Circuit Generation

Circuits are generated using Qiskit [**?**] and simulated with the Aer statevector simulator. Each circuit family is parameterized by qubit count n, number of operations $N_{\mathrm{ops}} = 1000$, and a reproducible master seed. For each $(n,\ \text{family})$ combination, $N_c = 1000$ unique circuits are sampled and measured with $s = 16n^2$ shots each. Table I summarizes the experimental sweep.

TABLE I
Experimental parameter sweep.

Parameter	Values
Circuit families	IQP, Clifford, Clifford+T
Qubit counts	4–20
Shot budget	$16n^2$ (quadratic)
Circuits per cell	1000
Operations per circuit	1000
Simulator	Qiskit Aer (statevector)

B. Shot Budget Design

A fundamental constraint in distribution estimation is that statistical features must be estimated from a finite number of samples. The estimation cost varies substantially across features:

a) Polynomial-cost features: The Hamming-weight histogram bins all 2^n outcomes into $n+1$ weight classes, reducing the estimation problem to a $(n+1)$-category multinomial. This requires $O(n/\varepsilon^2)$ shots for precision ε—linear in n. Single-qubit marginals and parity bias are single-qubit observables requiring $O(n/\varepsilon^2)$ and $O(1/\varepsilon^2)$ shots respectively. Nearest-neighbor connected ZZ correlations require $O(n^2/\varepsilon^2)$ shots across all pairs—quadratic in n.

b) Exponential-cost features: Shannon entropy $H = -\sum_x \hat{p}(x) \log_2 \hat{p}(x)$ requires $O(2^n/n)$ shots for accurate estimation from raw empirical frequencies, because the estimator is biased whenever outcomes are unobserved—which becomes the norm above $\sim$10 qubits. Collision probability $\mathcal{C} = \sum_x \hat{p}(x)^2$ requires $O(2^{n/2})$ shots via the birthday problem threshold; below this, the estimator systematically overestimates $\mathcal{C}$ because unseen outcomes contribute zero rather than their true p^2 value.

These exponential requirements would demand a sample budget growing faster than polynomially with n, negating the scalability motivation of the approach. We therefore exclude Shannon entropy and collision probability from our feature set and restrict attention to the polynomial-cost features.

c) Quadratic shot scaling: The most expensive retained feature is the ZZ correlation suite, which needs $O(n^2)$ shots. We therefore adopt $s = 16n^2$, which provides consistent statistical quality for all retained features across the full 4–20 qubit range. At $n = 4$ this yields 256 shots; at $n = 20$ it yields 6400 shots—well within the capability of near-term quantum hardware and classical simulation.

C. Feature Extraction

For each set of s measurement samples (counts dictionary), we extract the following four feature groups:

a) Hamming-weight histogram: The normalized histogram $h[w] = P(|x| = w)$ for $w = 0, \ldots, n$. This collapses all 2^n outcome probabilities into $n+1$ weight-class probabilities, and is well-estimated with $O(n)$ shots.

b) Single-qubit marginals: $m_i = P(x_i = 1)$ for $i = 0, \ldots, n-1$, estimated from empirical counts. These are 1-local observables requiring $O(n)$ shots for all n marginals.

c) Parity bias:

$$P_{\text{bias}} = P(\text{even parity}) - P(\text{odd parity}) \tag{1}$$

A single global scalar that is particularly informative for IQP circuits, which have a structured parity signature due to their diagonal-gate construction.

d) Connected ZZ correlations: Nearest-neighbor connected correlations

$$C_{i,i+1} = \langle Z_i Z_{i+1} \rangle - \langle Z_i \rangle \langle Z_{i+1} \rangle \tag{2}$$

for $i = 0, \ldots, n-2$, where $Z \in \{+1, -1\}$ encodes the measurement outcome. These 2-local observables capture pairwise correlation structure and require $O(n^2)$ shots across all $n-1$ pairs.

e) Feature vector: All feature groups are concatenated into a fixed-length feature vector. The feature vector length for an n-qubit system is

$$\underbrace{(n+1)}_{\text{HW hist}} + \underbrace{n}_{\text{marginals}} + \underbrace{1}_{\text{parity}} + \underbrace{(n-1)}_{ZZ} = 3n + 1. \tag{3}$$

This grows linearly with n, keeping the classifier input dimension tractable throughout the 4–20 qubit range.

D. Classification Pipeline

We evaluate four classifiers from scikit-learn [**?**]: Logistic Regression (L-BFGS, max_iter=1000), Decision Tree (unlimited depth), Gradient Boosting, and Extra Trees (200 estimators). Data are split 80/20 train/test with stratification by class label. Accuracy is reported as the fraction of correctly classified test samples.

E. Cross-Domain Experiment: Circuit Data to Real-World Observations

The primary motivation for classifying distributions is to identify real-world observations as belonging to a known distribution class, without requiring labeled real-world training data. To evaluate this capability, we construct a dataset of synthetic real-world observations by bootstrap-resampling from the circuit feature vectors.

Each synthetic *observation* is formed by drawing $N_s = 50$ circuit feature vectors with replacement and averaging them:

$$\mathbf{x}_{\text{obs}} = \frac{1}{N_s} \sum_{k=1}^{N_s} \mathbf{x}_{(k)}, \tag{4}$$

where each $\mathbf{x}_{(k)}$ is sampled uniformly from the circuit data for that family and qubit count. This simulates an observer collecting N_s independent samples from an unknown process and summarizing them as an average feature vector—a natural aggregation strategy when the underlying distribution class is the quantity of interest rather than individual outcomes.

We run three experiments:

1) **Circuit-only**: train and test on raw circuit data (80/20 split).
2) **Observation-only**: train and test on aggregated observation data (80/20 split).
3) **Cross**: train on raw circuit data, test on aggregated observation data. Labels are matched by distribution class.

The cross experiment is the key result: it measures whether a classifier trained solely on circuit-generated data can identify the distribution class of real-world aggregated observations, without any labeled real-world training examples.

IV. Results

A. Classification Accuracy vs. System Size

Figure **??** shows average accuracy as a function of qubit count.

[TODO: Describe scaling trend. Does accuracy degrade, plateau, or improve with qubit count? Is there a crossover point?]

B. Per-Family Confusion Analysis

[TODO: Include confusion matrix or per-class precision/recall. Which pairs of families are hardest to distinguish? Hypothesis: Clifford vs. Clifford+T is harder than IQP vs. either.]

C. Feature Importance

[TODO: Report feature importances from the Extra Trees or Gradient Boosting model. Which statistics are most discriminative?]

D. Cross-Domain Generalization: Circuit Training to Real-World Observations

Table **??** and Figure **??** summarize the three cross-domain experiments. **[TODO: Fill in numerical results once data is available.]**

A consistent pattern emerges across system sizes: tree-based methods (Decision Tree, Gradient Boosting, Extra Trees) dominate on both the circuit-only and observation-only tasks but fail to generalize in the cross experiment, while Logistic Regression remains the only classifier reliably above the random-chance baseline in the cross setting.

a) Interpretation: This result follows from the mechanics of the aggregation operation used to construct real-world observations. By the central limit theorem, averaging $N_s = 50$ feature vectors preserves class means but compresses feature variance by a factor of $1/\sqrt{N_s}$. Tree-based classifiers learn axis-aligned threshold rules of the form "$x_j > \tau$"; these absolute threshold values are calibrated to the variance of the training distribution and become misaligned when feature variance collapses under aggregation. Logistic Regression, by contrast, learns a global separating hyperplane $\mathbf{w}^\top \mathbf{x} > b$. Since the class means are preserved under averaging ($\mathbb{E}[\bar{\mathbf{x}}] = \mathbb{E}[\mathbf{x}]$), the classes remain on the correct sides of the learned hyperplane, and Logistic Regression generalizes.

This is not a failure of the tree-based methods; it is a statement about the nature of the distribution shift. The aggregation operation preserves the *linear* structure of feature space—the direction separating distribution classes—while destroying the fine-grained distributional structure (absolute feature scales, tail behavior, local density) that tree methods exploit. Logistic Regression's ability to generalize is therefore evidence that the class-separating signal is linearly accessible and robust to the aggregation transform.

b) Scaling behavior at high qubit counts: At 9–10 qubits, Logistic Regression's cross-domain accuracy decreases toward the random baseline, while tree methods begin to show marginal signal. We attribute this to shot-noise saturation: at $n = 10$ with $16 \times 100 = 1600$ shots, features such as parity bias and ZZ correlations become noisier due to shot statistics, degrading the class-separating direction that Logistic Regression relies on. Simultaneously, the variance compression from averaging becomes less dramatic relative to the existing shot noise, allowing tree methods to occasionally find splits that survive the distribution shift. The 11–20 qubit data (to be collected on HPC) will test whether this trend continues or reverses, and whether Logistic Regression remains the sole generalizing classifier at higher system sizes.

V. Discussion

A. Merit of Distribution Classification

The ability to classify observations by their underlying distribution class, rather than by raw feature values calibrated to a specific dataset, has broad practical implications. Many physical and engineered systems produce observables whose statistical structure encodes meaningful information about the generative process—yet obtaining labeled real-world training data for every deployment context is costly or infeasible.

The framework demonstrated here addresses this directly: by identifying the relevant distribution class in advance and generating labeled training data from that class via quantum circuits or classical simulation, a classifier can be deployed against real-world observations without any labeled field examples. The cross-domain experiment provides a proof of concept: the classifier is trained on circuit data alone and tested against aggregated observations constructed to resemble what a real-world observer would collect.

B. Mapping Real-World Quantities to Distribution Classes

[TODO: Discuss how real-world processes can be mapped to the distribution classes studied here. What physical or observational properties determine whether a process resembles IQP, Clifford, or Clifford+T distributions? Identify specific quantities that serve as analogues of qubit count (system complexity or dimension) and shot budget (observation or integration budget). Provide at least one concrete mapping to a domain of interest.]

The framework is not limited to three classes; additional distribution classes can be incorporated by extending the circuit family library or by including other generative models. The key requirement is that the class-specific distributional signatures be linearly separable in the chosen feature space—a condition that the cross-domain experiment directly tests.

C. Limitations

This study uses classical simulation of quantum circuits. Results reflect ideal (noiseless) quantum behavior; real device noise will alter the distributions and likely degrade classification accuracy. Extending to noisy simulation (Aer noise models) or real hardware is a natural next step.

The feature set is hand-engineered based on shot-scaling considerations. A learned representation (e.g., a neural network trained end-to-end on the counts dictionary) may extract more discriminative features, at the cost of interpretability and the tractability guarantees provided by the polynomial-cost feature set.

The synthetic real-world observations are constructed by bootstrap-resampling from circuit data rather than from a physically motivated generative model specific to any application domain. Validation against data from a concrete real-world system is required before claims of applicability to that domain can be made.

D. Path to Broader Deployment

[TODO: Discuss what it would take to apply this pipeline to a concrete real-world domain: (1) identify what quantity maps to system dimension and sample count, (2) establish which distribution class the real-world process belongs to, (3) generate sufficient labeled circuit data, (4) deploy the linear classifier. Note that the current pipeline is hardware-agnostic at the Qiskit level and the classifiers are lightweight at inference time.]

VI. Conclusion

We have presented a framework for classifying probability distributions using statistical fingerprinting, with quantum circuit families providing canonical, reproducible training data for distinct distribution classes. Three families—IQP, Clifford, and Clifford+T—are used to generate labeled samples across 4–20 qubits; classical machine-learning classifiers are then trained on these samples and evaluated both on held-out circuit data and on aggregated real-world observations constructed from the same underlying distributions.

A key methodological contribution is the identification of which features are tractable at scale: Shannon entropy and collision probability are excluded because they require exponential sample budgets, while Hamming-weight histogram, single-qubit marginals, parity bias, and connected ZZ correlations are retained because their estimation cost scales polynomially with system size. A quadratic shot budget ($s = 16n^2$) is adopted to match the worst-case cost of the ZZ feature suite.

The cross-domain experiment—training on circuit data and testing on aggregated real-world observations—reveals that Logistic Regression is the only classifier that generalizes across this distribution shift. This result is explained by the fact that aggregation preserves class means (and hence the linear separating structure of feature space) while destroying the absolute threshold structure that tree-based methods rely on. This finding provides a concrete and testable criterion for cross-domain applicability: the distribution classes must be linearly separable in the chosen feature space.

[TODO: State the best accuracy achieved and at what resource level.] The scaling behavior of classification accuracy with system size provides concrete guidance for the sample resources required to deploy this pipeline in practice. These results establish a quantitative foundation for applying distribution-level statistical fingerprinting to real-world classification problems and provide a reproducible baseline for future work incorporating noise models, hardware experiments, and alternative feature representations.

Acknowledgments

[TODO: Funding sources, HPC allocation, institutional acknowledgments.]